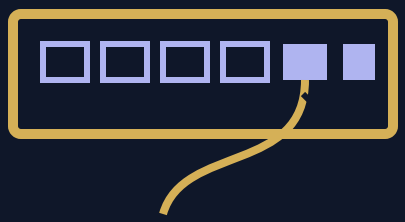
TEACHING CASE · EDUCATION · VULNERABILITY MANAGEMENT

Known doors stay open too long. Students as patch crew.

How a technical college can shrink its exposure with a free scanner, a weekly routine, and vendor access that expires, and turn its own students into part of the fix.



47 → 9 days

that a critical flaw stayed open on internet-facing servers

10 : 1

first-year return on about $6,990 of spending

21 of 24

students completed the graded lab with the lab box, vs. 14 with lecture

Anèl Henning · Logical Coders, Tampa, FL

Fictional organization · illustrative data

01 · THE WEAK SPOT

Two ways in: a vendor's door and an old server

SPRING 2023
A ransomware group exploits a known flaw in PaperCut print servers, targeting the education sector (CISA & FBI, 2023).

PATCH EXISTED
The fix was already available. The servers that were hit simply had not applied it.

DEC 2024
A stolen credential opens PowerSchool's customer-support portal, and school districts' data is copied (PowerSchool, 2025).

THIS CASE
A college closes both doors, and its students help keep them closed.

2,400	180	4	6	48
STUDENTS	STAFF	IT TEAM	ALWAYS-ON VENDORS	STUDENTS IN TRIAL

Colleges store Social Security numbers, financial aid records, and grades, and many depend on outside vendors to run key systems. Both recent incidents come down to time and access: how long a known hole stays open, and who can walk through a vendor's door.

NIST's patch guidance treats patching as routine maintenance that should be planned, not a fire drill (Souppaya & Scarfone, 2022). Pelican Bay Technical College is a fictional college with a cloud student information system (SIS), a print server in each building, and patching tracked in a spreadsheet.

ABOUT THIS CASE

Pelican Bay Technical College is fictional, and every trial number is illustrative. Students should never scan production systems without written authorization and supervision.

TWO GROUPS OF USERS

Following ISO 9241-210 (ISO, 2019), we interviewed both groups who would use the solution.

- **IT staff:** "We know what needs patching. We just can't find the time to check every server."
- **Students:** "Labs are slides and screenshots. I've never touched a real patch panel or run a scanner."
- **Define.** Let supervised students help run the weekly scan in a safe copy. IT gains hands, and students gain real experience.

02 · WHAT WE SET OUT TO PROVE

Three hypotheses, three trials

H1 · EXPOSURE

A weekly scan shrinks the window

A weekly scan plus a 14-day deadline for patching internet-facing systems will cut the days that critical flaws stay open.

SUPPORTED

H2 · VENDOR ACCESS

Access that expires

Time-limited, MFA-protected access approved per request will remove standing vendor accounts without slowing support (Joint Task Force, 2020).

SUPPORTED

H3 · LEARNING

Hands on beats slides

Students using the lab box and real scanning software will complete a graded lab more often and feel more capable than students taught by lecture.

SUPPORTED

FIG. 5 · PELICAN BAY TECHNICAL COLLEGE · VENDOR ACCESS + SCAN-AND-PATCH LOOP

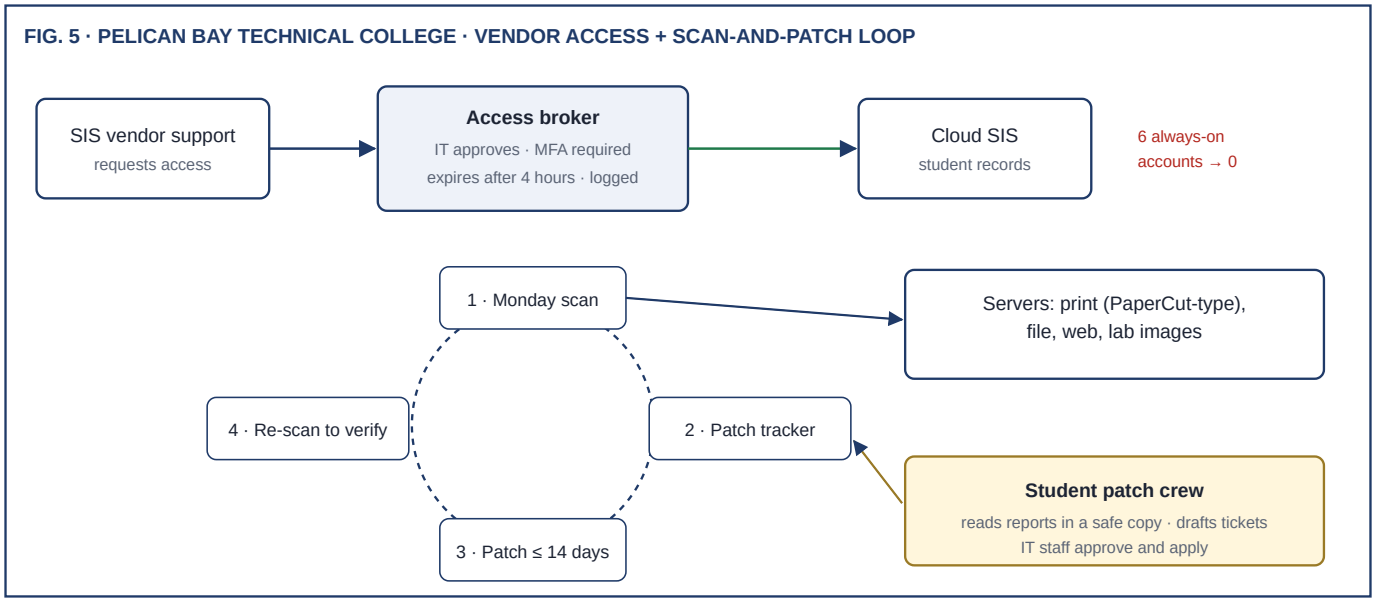


Figure 1. How the solution is built. Vendors no longer have standing accounts. They ask, IT approves, and access expires after four hours. A weekly loop scans every server, logs findings in a tracker, patches internet-facing systems within 14 days, and re-scans to confirm the fix. Supervised students help read reports and draft tickets.

TRIAL A · EXPOSURE WINDOW

We compared 10 weeks before the change with 10 weeks after. For each critical flaw on an internet-facing server, we measured the days from when a fix was available to when it was installed. A free scanner, Greenbone Community Edition, ran every Monday.

TRIAL B · VENDOR ACCESS

We counted always-on vendor accounts before and after, and timed how long vendors waited for approval.

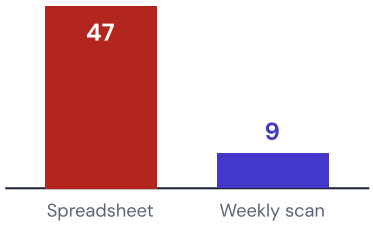
TRIAL C · CLASSROOM

Two sections of Network Fundamentals, 24 students each, took the same unit. One had lecture and screenshots. The other used the Patch Panel Lab Box and ran the scanner on a practice server inside it. Both did the same graded lab and took a self-efficacy survey before and after (Bandura, 2006).

03 · WHAT HAPPENED

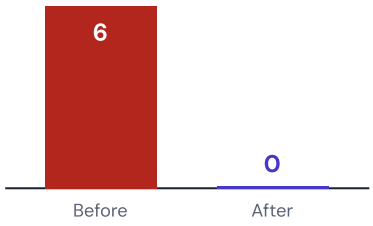
From seven weeks exposed to about one

Average days a critical flaw stayed open



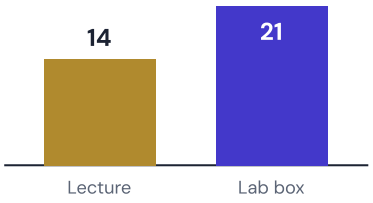
Test with Mann–Whitney on real data

Always-on vendor accounts



Median vendor wait for approval: 12 minutes

Completed the graded lab (of 24)



Fisher's exact, $p = .049$

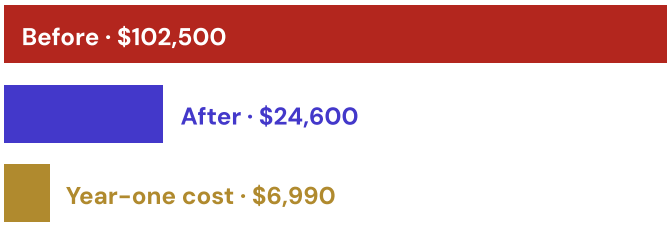
Illustrative data for a fictional college. Self-efficacy (1–5) rose from 2.5 to 3.9 with the lab box, and from 2.6 to 3.1 with lecture.

All three hypotheses are supported in the illustrative data. Standing vendor access dropped to zero, and vendors waited only a few minutes. In the classroom, the lab-box section completed the graded task more often, with a difference just under the .05 cutoff. The sections were intact classes, not randomly assigned students, so a real study should confirm it.

04 · MONEY AND TIME

A free scanner and a weekly habit

EXPECTED YEARLY LOSS (ALE)



ALE = single loss ($410,000) × yearly chance (0.25 → 0.06)

Building time-limited vendor access into the SIS contract at signing costs nothing extra. Adding it after a breach means renegotiating under pressure, and late fixes cost more (Boehm, 1981).

10 : 1

return in year one: $70,910 net benefit on $6,990 spent

Vendor access broker	$3,000
Instructor prep time	$1,350
12 lab boxes	$1,320
Scanner setup (free software)	$1,320
Total, year one	**$6,990**

Time saved: about 5 hours a week of manual checking, or 260 hours a year for a four-person team. The lab boxes are reused every term.

See the fix on screen, feel it on the panel

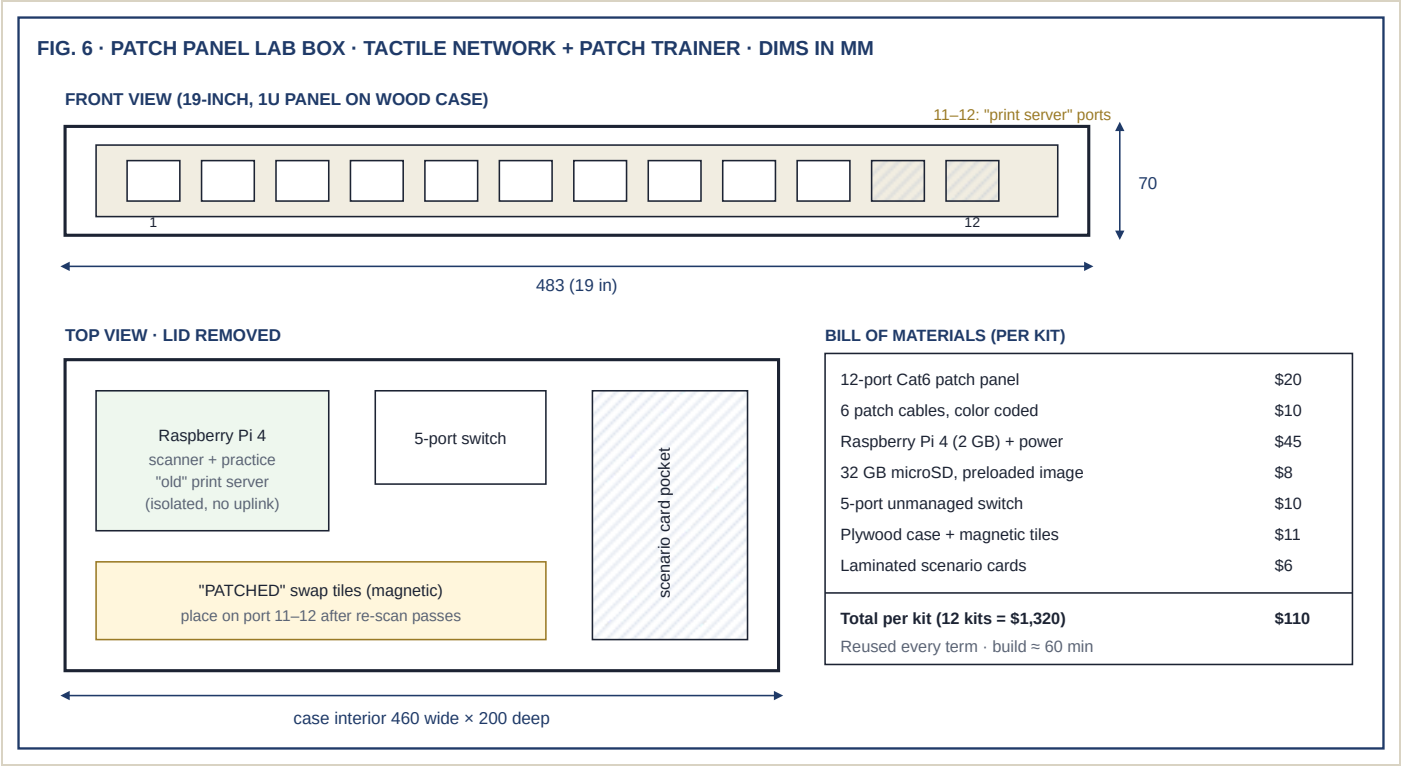


Figure 2. How the training kit is made. A real patch panel sits on the front of a wooden case. Inside, a Raspberry Pi runs the scanner and a deliberately outdated practice print server on an isolated network with no internet uplink. Students cable the "print server" ports, run the scan, apply the update, re-scan, and place a magnetic "PATCHED" tile on the port. The parts cost $110, and the box is reused every term.

THE PROBLEM TRAIL: ONE HABIT, FIVE STEPS

1

Notice

"Print jobs are slow and an alert names the print server."

2

Guess

"Is it running an old version with a known flaw?"

3

Test in software

Scan the practice server, find the critical finding, and look it up in CISA's catalog.

SOFTWARE

4

Prove by hand

Trace the cable to ports 11–12, update, re-scan, then place the PATCHED tile.

TACTILE

5

Report

Write a plain-language patch ticket: what was wrong, the risk, the fix, and the check.

BUILDING SELF-EFFICACY ON PURPOSE

Self-efficacy, the belief that you can succeed at a task, predicts effort and persistence (Bandura, 1977, 1997). Each lab is built around Bandura's four sources.

Mastery

Students succeed at a real task: the finding clears on re-scan.

Seeing others

A peer goes first while the group watches.

Encouragement

The instructor gives specific feedback on each ticket.

Calm body

A closed box where nothing real can break.

WHY HANDS-ON LEARNING WORKS

The trail follows problem-based learning (Hmelo-Silver, 2004) and Kolb's (1984) cycle of experience, reflection, and practice. Moving from cable to screen and back ties the physical network to the abstract idea of a vulnerability (Wilson, 2002), and active designs like this consistently outperform lecture alone (Freeman et al., 2014). Students who finish the unit can join the supervised patch crew, which gives them résumé experience and gives IT more capacity.

> *"They see the fix on screen and feel it on the panel."*
>
> TRIAL C · 21 OF 24 COMPLETED THE LAB, VS. 14 OF 24

TAKEAWAYS FOR AUDITORS

- Ask how many vendor accounts are always on, and who approves each session.
- Pull the last 10 weeks of scan reports and measure days-to-patch yourself.
- Check that every fix was confirmed by a re-scan, not just a ticket closure.

LIMITATIONS

The college and all results are fictional. The classroom comparison used two intact sections, not randomly assigned students, so instructor or class-time differences could affect results. A real study should randomize where possible, use one instructor, and get IRB approval before collecting student data.

CONCLUSION

The PowerSchool and PaperCut incidents share a lesson: known doors stay open too long. A technical college can close them with tools it already has. By making students part of the patch crew, it gets more security work done and gives students something no slide can: the experience of fixing a real problem with their own hands.

REFERENCES

Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review, 84*(2), 191–215. https://doi.org/10.1037/0033-295X.84.2.191

Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman.

Bandura, A. (2006). Guide for constructing self-efficacy scales. In F. Pajares & T. Urdan (Eds.), *Self-efficacy beliefs of adolescents* (pp. 307–337). Information Age Publishing.

Boehm, B. W. (1981). *Software engineering economics*. Prentice-Hall.

Cybersecurity and Infrastructure Security Agency & Federal Bureau of Investigation. (2023, May 11). *Malicious actors exploit CVE-2023-27350 in PaperCut MF and NG* (AA23-131A). https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-131a

Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *PNAS, 111*(23), 8410–8415. https://doi.org/10.1073/pnas.1319030111

Hmelo-Silver, C. E. (2004). Problem-based learning: What and how do students learn? *Educational Psychology Review, 16*(3), 235–266. https://doi.org/10.1023/B:EDPR.0000034022.16470.f3

International Organization for Standardization. (2019). *Ergonomics of human-system interaction — Part 210* (ISO 9241-210:2019).

Joint Task Force. (2020). *Security and privacy controls for information systems and organizations* (NIST SP 800-53, Rev. 5). https://doi.org/10.6028/NIST.SP.800-53r5

Kolb, D. A. (1984). *Experiential learning*. Prentice-Hall.

PowerSchool. (2025). *Cybersecurity incident* [Notice]. https://www.powerschool.com/security/sis-incident/

Souppaya, M., & Scarfone, K. (2022). *Guide to enterprise patch management planning* (NIST SP 800-40, Rev. 4). https://doi.org/10.6028/NIST.SP.800-40r4

Wilson, M. (2002). Six views of embodied cognition. *Psychonomic Bulletin & Review, 9*(4), 625–636. https://doi.org/10.3758/BF03196322